

DEDEN HIDAYAT

SOFTWARE ENGINEER - FULL-STACK, IOT, INDUSTRIAL SYSTEMS

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SUMMARY

Software Engineer bridging industrial hardware and cloud software. Hands-on across embedded firmware, 10+ industrial protocols (Modbus, BACnet, SNMP, OPC-UA, MQTT), Python middleware, and Next.js/TypeScript dashboards. Currently in R&D at PT GSPE, building protocol middleware and real-time monitoring platforms, including SMARTRACK, a full-stack data-center management system with live IoT telemetry.

SKILLS

Languages	TypeScript, JavaScript, Python, C / C++, Dart
Frontend	Next.js, React, Vue 3, Tailwind CSS, Framer Motion
Backend	Node.js, Flask, Prisma, PostgreSQL, Supabase, REST & WebSocket APIs
IoT / Protocols	MQTT, Modbus RTU/TCP, BACnet/IP, SNMP, OPC-UA, EtherNet/IP, IEC 61850, Sparkplug B
Embedded	ESP32, Raspberry Pi, ESP-IDF, Arduino
Tools / Infra	Docker, Git, Linux, ROS2, Vercel

EXPERIENCE

R&D Engineer PT GRAHA SUMBER PRIMA ELEKTRONIK (GSPE)	2023 - Present
- Architect middleware for 10+ industrial protocols, bridging bare-metal firmware to cloud telemetry. - Built real-time telemetry pipelines publishing device data to MQTT with bidirectional reverse-flow control. - Developed SMARTRACK, a full-stack data-center management platform with live monitoring, AI analytics, and RBAC security. - Delivered AI vision and STT/TTS systems for production monitoring and automated reporting.	
Freelance Engineer INDEPENDENT	2023 - Present
- Built IoT and web projects for clients, including real-time dashboards, protocol bridges, and mobile control apps. - Delivered end-to-end systems across firmware, middleware, APIs, and frontend interfaces.	

SELECTED PROJECTS

SMARTRACK - Data-Center Management Platform FULL-STACK / DCIM Full-stack DCIM platform with real-time MQTT/WebSocket monitoring, multi-protocol IoT ingest (Modbus, SNMP), AI analytics, drag-and-drop automation, RBAC, AES-256 security, PUE tracking, and carbon tracking. Next.js, TypeScript, Prisma, PostgreSQL, MQTT, Docker
Industrial Telemetry & Protocol Bridge FIRMWARE / MIDDLEWARE Real-time poller for industrial devices using Modbus and SNMP, publishing normalized telemetry to MQTT with bidirectional reverse flow on Raspberry Pi and ESP32 nodes. Python, MQTT, Modbus, SNMP, Raspberry Pi, ESP32
AGV Simulation & Web Control ROBOTICS ROS2 and Gazebo autonomous guided vehicle with Vue 3 control UI, Nav2 navigation, SLAM mapping, auto-docking, and keepout zones. ROS2, Gazebo, Nav2, SLAM, Vue 3, Docker
IoT BOM List v2 FULL-STACK / INFRASTRUCTURE BOM and inventory platform for IoT hardware teams, covering component databases, PCB assemblies, supplier pricing, Excel import/export, audit logs, backups, and Docker/nginx deployment. Next.js, TypeScript, Hono, Prisma, PostgreSQL, Docker, nginx
Production Machine Monitor (AI Vision) DEPLOYED SYSTEM Real-deployed monitor for CNC and laser machines using a C# screenshot agent and a Flask server running Qwen2.5VL locally for automated daily production reports. Python, Flask, C#, Ollama, Qwen2.5VL

EDUCATION

Teknik Elektronika Industri SMK NEGERI JAWA TENGAH	2020 - 2023
Industrial electronics foundations, including microcontrollers, electrical systems, and hardware-software integration.	